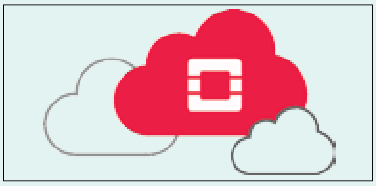
OpenStack releases Yoga, a new IaaS cloud

The OpenStack community has released Yoga, the 25th version of open source cloud infrastructure software. Yoga highlights include support for advanced hardware features such as SmartNIC DPUs, improved integration with cloud-native software such as Kubernetes and Prometheus, and reduction of technical debt to maintain a stable and reliable OpenStack core.

OpenStack, the open infrastructure-as-a-service standard, is the one infrastructure platform for deployment of diverse architectures—bare metal, virtual machines (VMs), graphics processing units (GPUs) and containers. It is one of the most active open source projects in the world, supported by a vibrant and engaged community of developers globally. Over the span of just 25 weeks, almost 13,500 changes authored by over 680 contributors from more than 125 organisations and 44 countries were merged into the Yoga release.



The feature advancements in the new release include hardware enablement extended for SmartNIC DPUs. Local IP is added to Neutron, focused on high efficiency and performance of the networking data plane for very large scale clouds, or clouds with high network throughput demands.

Some other basic changes include that Ironic, OpenStack's bare metal provisioning program default deployment boot mode, now supports UEFI instead of legacy BIOS. Meanwhile, Kolla is depreciating binary images. In the next release, all binary image support will be removed. Users must migrate to source-based images going forward.

The MHKiT team uses the MATLAB platform to import a large amount of data from a fleet of buoys, such as wave heights, current speeds, and ocean temperatures. The team has also included historical wave data to help predict the weather and climates that marine energy technology may encounter offshore. This so-called hindcast data set will eventually encompass the whole of United States.

MHKiT is a popular tool that has been downloaded over 4,000 times in just four years (although some of those downloads are repeat customers coming back for more).

Google and Linux Foundation release Nephio for telecom market

The Linux Foundation has extended its push to bring open source projects into the telecom market by partnering with Google Cloud to develop the Kubernetes based Nephio project. This project aims to make it easier for telecom operators to deploy and manage multi-vendor cloud infrastructure and network operations across large-scale edge deployments by providing Kubernetes based cloud-native intent automation and automation templates.

Nephio sits on top of a Kubernetes substrate, either directly handled by an operator or via a hyperscaler-based platform such as Google Config Connector, Amazon Web Services (AWS) Kubernetes Controllers, and Azure Service Operator.

Kubernetes' cloud-native orchestration capability is used to enable operators to roll out and manage new services in their 5G and edge deployments. This will allow for a speedier on-boarding of network functions into a production environment, similar to how hyperscalers employ the DevOps process.

"We believe that true cloud nativeness lies in the fact that Kubernetes can actually automate all the way down from describing a network function or a service to the infrastructure including all the underlying networking components," stated Gabriele Di Piazza, senior director of product management for telecom at Google Cloud.

The project is open about the fact that it does not provide end-to-end service orchestration, cloud infrastructure, or network operations.

The project's largest hurdle appears to be providing a "carrier-grade" option for telecom operators as they build up their 5G networks. This has long been an issue since telecom operators compete in a highly regulated market with government and public safety regulations that often exceed those of traditional cloud based operators.

Oracle makes Solaris 11.4 CBE available

Oracle has started making a new version of Solaris. Oracle Solaris 11.4 CBE was announced in early March this year, and is the 'Common Build Environment' for open source developers and strictly non-production personal use, which is required if you want Solaris for new installations after 2022.

The new Solaris 11.4 CBE is effectively a rolling release, and Oracle hopes that by doing so, it will make it easier to integrate the open source software that Solaris relies on, rather than being locked onto the dated 11.4.0 GA release.

An Oracle account is required to download the new Solaris 11.4 CBE. The CBE builds are sometimes defined as "pre-release builds of a particular SRU, analogous to a beta." The Oracle Technology Network Early Adopter License Agreement for Oracle Solaris grants the non-production usage licence.

Under Oracle support contracts, users will be able to upgrade from these free CBE releases to paid SRU releases. The Oracle Solaris blog has more information for people interested in Oracle Solaris 11.4 CBE.